Joint-Embedding Predictive Architectures and Latent Residual Conditioning for Multivariate Time-Series Anomaly Detection

Andre Williams, Imadeldin Mahgoub, *Senior Member, IEEE,* and Hamid Akbarian

***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize observation-space error, fitting stochastic sensor noise that produces heavy-tailed residuals and severe false-alarm flooding under automated threshold calibration. We investigate how Joint-Embedding Predictive Architectures (JEPAs) condition predictive residuals in a 32-dimensional latent space under distribution shift, examining unconstrained predictors, Operator-Entropy JEPA (OpEntropy JEPA) with spectral conditioning, and Reynolds-Stress JEPA with a Cayley-accretive Crouzeix linear core and Stiefel covariance closure. Across 33 continuous telemetry streams from seven physical datasets (19 non-Gasoil Heating Loop (GHL) streams at five seeds and 14 GHL streams at three seeds) under chronological held-out Streaming Peaks-Over-Threshold (SPOT) calibration, we evaluate stream-macro realized false-positive rates (FPRs) and out-of-sample calibration error from the 0.1% design risk as primary outcomes. Our central contribution is relative false-alarm suppression and drift reduction rather than attainment of the nominal 0.1% target or universal detection superiority. Latent scoring lowers unweighted stream-macro held-out FPRs three-fold under retrospective buffered attribution (9.3% to 10.9% for base latent predictors versus 30.0% to 30.7% for TimesNet and TranAD across the 19 non-GHL streams; 5.5% to 6.4% versus 19.6% to 19.9% stream-macro across all 33 streams) and eight- to ten-fold under strictly causal streaming (1.9% to 2.8% versus 22.2% to 24.3% across 19 non-GHL streams; 1.5% to 2.0% versus 14.2% to 16.1% stream-macro across all 33 streams), while point F1 and precision-recall ranking remain comparable to baselines. Matched-encoder factorial ablations show that Layer Normalization (LayerNorm) bottlenecks compress residual kurtosis, whereas learned latent projections halve calibration-to-holdout drift relative to normalized observation forecasters. Equivalence testing confirms that Koopman spectral regularization conditions operator spectra without altering point F1 within a 0.015 to 0.030 margin, while short prediction horizons (one to four steps) eliminate structural emission latency and hybrid sensor variance monitors recover quiet single-channel faults at a quantified false-alarm cost.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Threshold calibration, Extreme value theory, Representation learning.**

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I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable healthcare [1], [2]. In NASA SMAP and MSL rovers, industrial process loops, and biometric sensor arrays, dozens of channels track power buses, thermal gradients, vibrations, and kinematics [3]. Monitors must identify subsystem faults without overwhelming operators with nuisance alerts.

False-Alarm Flooding Under Automated Threshold Calibration: Operational monitoring requires automated decision rules that convert continuous scores into binary alarms using nominal calibration data alone. Prevailing deep detectors (TimesNet, TranAD, Anomaly Transformer) minimize point-level reconstruction error in raw observation space [4]–[6]. Because raw telemetry contains sensor jitter and scale disparities, observation residuals exhibit heavy-tailed excess kurtosis ($\gamma_2 > 8$) and serial clustering. When paired with Streaming Peaks-Over-Threshold (SPOT) [7] at design risk $q = 10^{-3}$ (0.1%), heavy-tailed residuals suffer calibration breakdown on unseen nominal data, triggering false alarms across 22%–31% of nominal timesteps.

Latent Residual Conditioning and Scope of Claims: Predicting the *summary* of a future window, rather than every raw sample, ignores unforecastable sensor jitter. We study Joint-Embedding Predictive Architectures (JEPAs) [8], which forecast a 32-dimensional latent summary of the next S steps ($S = 64$ by default, or $S \in \{1, 4\}$ for low-latency tripwires) from the preceding 256, without synthetic anomaly injection [9]. Beyond unconstrained predictors (`TS-JEPA`), we formulate physically grounded variants: an *Operator-Entropy JEPA* (OpEntropy JEPA) regularized by Koopman spectral entropy and a Bregman energy barrier, and a *Reynolds-Stress JEPA* enforcing Cayley-accretive Crouzeix contractivity and Stiefel positive semidefinite second-moment closure. We evaluate these structures as geometric conditioning priors via equivalence tests rather than claims of detection superiority. Our contribution is relative false-alarm suppression and reduced calibration-to-holdout drift rather than solved absolute calibration at $q = 10^{-3}$. We evaluate: (i) how much latent residuals reduce realized false-positive rates (FPRs) and out-of-sample calibration error $|\text{FPR}_{\text{hold}} - q|$ under static, declustered, and rolling calibrators; (ii) how normalization, learned latent projection, and horizon isolate the mechanism of improvement; (iii) how detection varies across abrupt, gradual, and quiet

single-channel faults (and the false-alarm cost of hybrid sensor monitors); and (iv) the trade-off between score-emission latency ($S-1$) and detection quality.

A. Contributions

1) **Factorial isolation of tail compression vs. drift reduction:** Across 14 matched-encoder and non-parametric arms (269k–307k online parameters), a layer-normalization (LayerNorm) bottleneck bounds heavy residual tails (γ_2 : $6.10 \rightarrow 0.91$), whereas a learned latent projection halves calibration-to-holdout drift (D_{KS} : $0.376 \rightarrow 0.193$–0.214) relative to normalized observation forecasters (Section VI-E, Table II(c)).
2) **Fewer false alarms, comparable detection, and hybrid quiet-fault recovery:** On 33 streams from seven datasets, latent scoring cuts held-out FPRs and calibration error three-fold under retrospective (buffered) attribution and eight- to ten-fold under strictly causal streaming, and nearly doubles event recall at a 1% realized FPR budget. Point-wise F1 (Point-F1) and precision-recall area under the curve (PR-AUC) remain comparable to baselines, and pairing latent prediction with a sensor variance monitor raises quiet single-channel event recall from 0.19–0.28 to 0.42–0.50 at a $+15.1\%$ nominal FPR cost (Table II(c), Table III, Figs. 3(b) and 4(b)).
3) **Verified operator conditioning and short-horizon latency control:** Proposition 1 proves Koopman non-degeneracy across the full objective (including the discrete Lyapunov covariance defect, repulsive origin curvature at $\mathbf{M} = \mathbf{0}$, and finite-ϵ energy bounds), while the Cayley-accretive Crouzeix linear core and sign-stabilized QR Stiefel retractions enforce contractive linear transport and interior covariance closure. Equivalence tests confirm that Koopman spectral regularization leaves Point-F1 unchanged within $\Delta = \pm 0.015$–0.030 while shortening detection delay, and short horizons ($S \in \{1, 4\}$) eliminate the 63-step emission lag (Tables III and II, Fig. 4(a)).
4) **Decomposed calibration error across static, declustered, and adaptive rules:** A refitted permutation decomposition estimates point-estimate shares of 56.6%–64.1% for inter-block drift in pooled excess JEPA risk above $q = 10^{-3}$ (77.5%–92.3% with resolved positive confidence intervals (CIs) for reconstructors, whereas per-stream JEPA CIs include zero across 11 zero-FPR streams), and runs-declustered SPOT and rolling quantiles narrow the gap on step and variance shifts (Table IV).

Roadmap. Section II reviews related work and Section III details the benchmark cohort. Section IV formulates the architectures, Section V defines the protocol, Sections VI and VII report ablations and benchmark results, and Section VIII concludes.

TABLE I
KEY OPERATIONAL AND STATISTICAL TERMS USED THROUGHOUT THE MANUSCRIPT.

Term	Meaning
Latent residual	Gap between the predicted and observed 32-dimensional summary of the next window.
SPOT	Streaming extreme-value rule that sets an alarm threshold from nominal scores alone.
Calibration error	Absolute gap $\lvert \mathrm{FPR}_{\mathrm{hold}} - q \rvert$ between realized held-out nominal FPR and design risk $q{=}10^{-3}$.
Event recall	Fraction of true anomaly segments with at least one alarm.
Buffered	Retrospective scoring that spreads each window score over its S steps.
Causal	Scoring that uses only past data and assigns each score to its final step $\tau = t + S - 1$.
Calibration drift	Shift between calibration and later nominal residuals, measured by Kolmogorov–Smirnov distance.
Excess kurtosis	Tail heaviness of the residual distribution; large values mean rare, extreme spikes.
Equivalence test	Test that shows two models differ by less than a stated margin (Section V).

II. RELATED WORK

Observation-Space Forecasting and Reconstructive Detectors: Deep time-series anomaly detectors predominantly forecast sensor readings or reconstruct input windows in raw observation space, flagging timesteps where point-level error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ exceeds a threshold. Autoregressive long short-term memory (LSTM) networks with non-parametric dynamic thresholding [3] model sequential dependencies directly on spacecraft channels. Recent Transformer and spectral reconstructors introduce specialized attention and frequency modules: TranAD [6] combines adversarial two-phase training with self-conditioning sub-series attention; Anomaly Transformer [4] contrasts prior Gaussian kernel associations against series-level self-attention; and TimesNet [5] folds 1D sequences into 2D tensors along dominant Fast Fourier Transform (FFT) periods for 2D convolutional modeling. Because point-level losses penalize error across every channel equally, reconstructors fit stochastic sensor jitter and environmental fluctuations. In high-dimensional streams ($K \geq 25$), summing raw channel errors accumulates heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) residual spikes that destabilize extreme-value threshold fitting under baseline shift [10].

Contrastive and Joint-Embedding Predictive Architectures: Self-supervised representation methods map windows into a compact latent space. Contrastive detectors construct positive and negative pairs: TS2Vec [11] maximizes contextual agreement across hierarchical sub-series; Neural Contextual Anomaly Detection (NCAD) [9] contrasts full-window and context-only embeddings using injected synthetic anomalies; and DCdetector [12] employs dual patch-wise and in-patch attention without reconstruction. However, contrastive detectors depend on whether synthetic perturbations or patch discrepancies match test-time fault morphology (0.058 and 0.070 macro Point-F1 for NCAD and DCdetector vs. 0.160 for latent prediction across 11 datasets, Table III(d)). Joint-Embedding Predictive Architectures (JEPAs) [8], [13], [14] eliminate observation decoding and negative pairs by encoding context z_{ctx} and predicting the target embedding z_{tgt} produced by a separate target encoder, avoiding collapse via exponential moving average (EMA) target updates [15] or variance-covariance penalties such as VICReg [16], Barlow Twins [17], and batch whitening [18]. How latent predictive residuals behave under unsupervised extreme-value threshold calibration remains unexplored.

Koopman Operator Theory and Physics-Inspired Priors: Multi-sensor telemetry reflects physical dynamical trajectories [19]. Neural ordinary differential equations (ODEs) [20], Physics-Informed Neural Networks (PINNs) [21], and Hamil-

tonian or Lagrangian networks [22], [23] enforce differential conservation laws on physical coordinates, but require known governing equations and suffer from stiff gradient pathologies on uncalibrated sensor arrays [24]. Koopman operator theory [25], [26] lifts nonlinear dynamics into an observable space governed by a linear transition matrix $\mathbf{K}$ [27]–[29]. We regularize the singular value spectrum of $\mathbf{K}$ via Von Neumann spectral entropy and a Bregman energy barrier, distinguishing *physics-inspired* algebraic priors on abstract latent coordinates from *physics-informed* conservation laws and testing whether operator conditioning alters downstream detection via equivalence tests.

Extreme-Value Threshold Calibration and Evaluation Rigor: Streaming Peaks-Over-Threshold (SPOT) [7] fits a Generalized Pareto Distribution (GPD) to score exceedances above quantile u to extrapolate a threshold at design risk q. However, continuous telemetry violates the classical independent and identically distributed (i.i.d.) assumption through short-range serial correlation (quantified by the Ferro–Segers extremal index [30]) and inter-block regime drift. Because "point-adjusted" F1 severely inflates scores [10], [31], [32], we evaluate all models using unadjusted Point-F1, PR-AUC, causal vs. buffered attribution, and matched realized false-alarm operating curves.

III. Multi-Sensor Internet-of-Things and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams from NASA spacecraft archives [3], the University of California, Irvine (UCI) Machine Learning Repository [33], [34], TimeEval [10], and mTSBench [35]: 33 streams are evaluated under the primary multi-seed held-out protocol and 12 are reserved with explicit inclusion accounting below. Streams span four physical domains and distinguish naturally occurring operational or physiological anomalies from testbed or simulated faults: *spacecraft and solar telemetry* (six SMAP, K=25, 7 anomaly events; four primary and two short-nominal MSL entities, K=55, 6 primary events; and SWAN-SF solar magnetograms [36], K=38, 6,615 flare events, all natural operational faults); *wearable kinematics* (six Daphnet Freezing-of-Gait (FoG) streams [33] at 64 Hz, K=9, 72 natural gait-freezing episodes, where S=64 is a 1 s horizon); *industrial processes* (14 Gasoil Heating Loop (GHL) streams, K=16, 26 testbed valve/heater faults; the Genesis pick-and-place testbed, K=18, 2 actuator faults; and GECCO water chemistry at 1-minute resolution, K=9, 37 natural hydrological anomalies); and *smart infrastructure* (Metro I-94 hourly highway traffic, K=5, 25 natural traffic events; Room-Occupancy sensing [37], K=5; CalIt2 building flow [34], K=2; and six CICIDS network streams, K=72, simulated attack flows). Separately, Cohort 3 (Section VI-C) injects six controlled synthetic single-channel faults into six real streams.

Each stream provides nominal training sequences ($N_{\text{tr}} \in [2{,}074, 120{,}000]$ across the 33 primary streams) and labeled anomaly test sequences ($N_{\text{test}} \in [2{,}158, 114{,}001]$); channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatlines, matching standardized archive subsets in prior benchmarks [3], [10].

Each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by an unobserved 64-step isolation gap (Section V-A). Inclusion in the primary multi-seed protocol required two prespecified criteria: (i) $N_{\text{cal}} \geq 500$ calibration windows ($M_{\text{val}} = \lceil 0.35\, T_{\text{nom}} \rceil - 319 \geq 1{,}000$, i.e., $T_{\text{nom}} \geq 3{,}769$), and (ii) tractable stride-1 multi-seed runtime ($K \leq 55$, $N_{\text{test}} \leq 1.2 \times 10^5$). Exactly 33 streams across seven datasets satisfy both criteria (Table III(c)). All 12 reserved streams are accounted for: five fail $N_{\text{cal}} \geq 500$ (CalIt2 `traffic`, T_{nom}=940, N_{cal}=4; MSL `C-2`, T_{nom}=764, N_{cal}=0; MSL `D-16`, T_{nom}=1,451, N_{cal}=94; Room-Occupancy `default`, T_{nom}=1,073, N_{cal}=28; and `1`, T_{nom}=1,492, N_{cal}=101); six network streams exceed the stride-1 multi-seed runtime envelope (CICIDS `0,2,3,5,6,7`, K=72, $N_{\text{test}} \in [1.36 \times 10^5, 4.15 \times 10^5]$); and GECCO `water_quality` ($K$=9, 1 held-out event) is used in Cohorts 3–4. All 11 datasets and 130 archive streams appear in the secondary single-seed cross-check (Table III(d)).

Two caveats bound external validity: (i) SMAP and MSL contain documented labeling flaws [32]; and (ii) streams within SMAP, MSL, Daphnet, and GHL share hardware platforms (N=7 dataset clusters). We therefore report Dataset-Macro means, between-stream and within-stream seed standard deviations (SDs), zero-tuning leave-one-dataset-out (LODO) sensitivity, and dataset-clustered permutation tests (Table III(c)).

IV. Methodology: Operator-Entropy and Reynolds-Stress Joint-Embedding Predictive Architectures

A. Problem Formulation and Latent Space Prediction

Consider a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T \times K}$ recording K sensor channels over T timesteps. As shown in Fig. 1 (Stage 1), a sliding window of length $W = C+S = 320$ advances at stride 1 and splits the series at index t into: (i) a *context window* $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t-C : t, :] \in \mathbb{R}^{C \times K}$ ($C = 256$), establishing the nominal regime; and (ii) a *suspect target horizon* $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t+S, :] \in \mathbb{R}^{S \times K}$ ($S = 64$, up to decision timestamp $\tau = t + S - 1$), inspected for trajectory departures.

Unlike reconstructive detectors that minimize pointwise observation error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ and inherit leptokurtic sensor noise (Sections I and II), we shift prediction into a 32-dimensional latent space ($D = 32$, Fig. 1) using a Joint-Embedding Predictive Architecture (JEPA) [8]. An online context encoder E_θ maps $\mathbf{x}_{\text{ctx}}$ into $z_{\text{ctx}} \in \mathbb{R}^D$, while a target encoder E_ϕ maps $\mathbf{x}_{\text{tgt}}$ into $z_{\text{tgt}} \in \mathbb{R}^D$. To prevent constant-vector collapse, target weights ϕ are updated as an EMA of online weights θ, $\phi_k \leftarrow m_k \phi_{k-1} + (1 - m_k)\theta_k$ with cosine momentum $m_k \in [0.996, 0.9995]$ [13], [15].

B. Latent Predictor Architectures: Operator-Entropy and Reynolds-Stress JEPA

We evaluate three nested latent predictors sharing the same multiscale encoder backbone and coordinate saliency gate: an

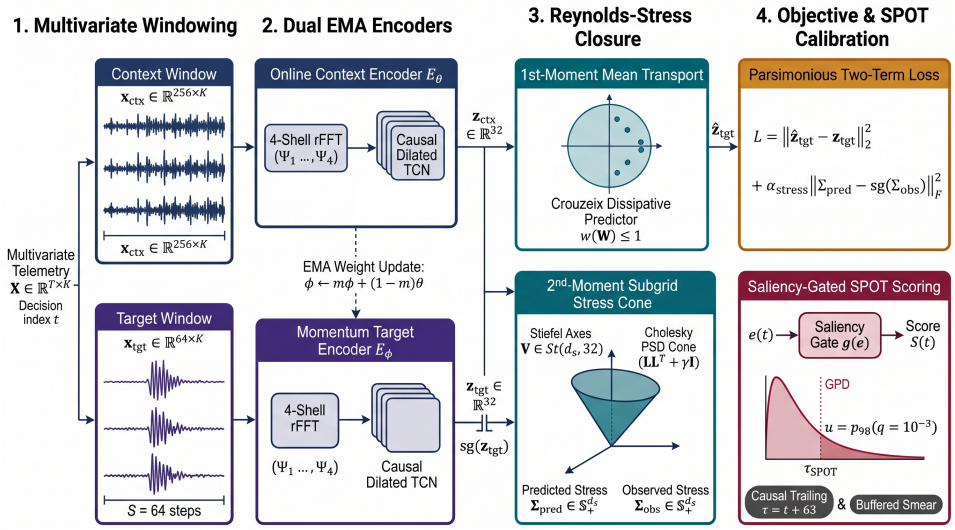


Fig. 1. Four-stage `Reynolds-Stress JEPA` architecture. (1) Multiscale windowing splits telemetry at index t into context ($\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{256\times K}$) and target ($\mathbf{x}_{\text{tgt}} \in \mathbb{R}^{64\times K}$) blocks processed by a 4-shell dyadic real fast Fourier transform (rFFT) front-end and causal dilated temporal convolutional network (TCN). (2) Dual online (E_θ) and exponential moving average (EMA) target (E_ϕ) encoders map windows into $\mathbb{R}^{32}$. (3) Reynolds-stress closure forecasts first-moment transport ($\hat{\mathbf{z}}_{\text{tgt}}$) via a Cayley-accretive Crouzeix predictor ($W(\mathbf{W}_{\text{Crouzeix}}) \subset \mathbb{D}$) and second-moment subgrid covariance ($\mathbf{\Sigma}_{\text{pred}}, \mathbf{\Sigma}_{\text{obs}} \in \mathcal{S}_+^{d_s}$, d_s=16) on a sign-stabilized QR Stiefel frame $\mathbf{V} \in \text{St}(d_s, 32)$. (4) Parsimonious two-term training objective and saliency-gated SPOT calibration under causal trailing ($\tau = t + 63$) and buffered smear mappings.

unconstrained multi-layer perceptron (`TS-JEPA`), a spectral-regularized Koopman transition (`Operator-Entropy JEPA`), and a structurally constrained second-moment closure model (`Reynolds-Stress JEPA`).

1) Operator-Entropy JEPA (Spectral Conditioning): **Formulation.** Koopman operator theory [25], [26] represents nonlinear dynamics as a linear map on learned observables, $\hat{z}_{\text{tgt}} = \mathbf{K} z_{\text{ctx}}$ with $\mathbf{K} \in \mathbb{R}^{D\times D}$. Trained on prediction error alone, $\mathbf{K}$ can suffer rank or norm collapse. We regularize both the shape and scale of its singular values $\sigma_1 \geq \cdots \geq \sigma_D \geq 0$ via $\mathcal{R}_{\text{spectral}}(\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K}) + \beta_{\text{norm}}\Phi_{\text{energy}}(\mathbf{K})$. Normalizing $\lambda_i = \sigma_i^2$ into $p_i = (\lambda_i + \epsilon_{\log})/\sum_{j=1}^{D}(\lambda_j + \epsilon_{\log})$ ($\epsilon_{\log} = 10^{-12}$) defines the Von Neumann spectral entropy $\mathcal{H}_{\text{spectral}}(\mathbf{K}) = -\sum_{i=1}^{D} p_i \log p_i$ and the *Spectral Entropy Deficiency* relative to uniform $\mathbf{u} = [1/D, \ldots, 1/D]^\top$:

$$\Delta\mathcal{H}(\mathbf{K}) = \log D - \mathcal{H}_{\text{spectral}}(\mathbf{K}) = \mathcal{D}_{\text{KL}}(\mathbf{p} \,\|\, \mathbf{u}) \geq 0. \quad (1)$$

Because $\Delta\mathcal{H}(c\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K})$ is scale-invariant for $c > 0$, a convex Bregman barrier on the energy ratio $r(\mathbf{K}) = \|\mathbf{K}\|_F^2/D$ prevents norm decay ($\epsilon = 10^{-6}$):

$$\Phi_{\text{energy}}(\mathbf{K}) = r(\mathbf{K}) - 1 - \log\big(r(\mathbf{K}) + \epsilon\big) + \log(1+\epsilon) \geq 0. \quad (2)$$

Combining both terms yields the training objective:

$$\mathcal{L}_{\text{Op-Entropy}} = \frac{1}{BD}\sum_{b=1}^{B}\big\|\mathbf{K} z_{\text{ctx}}^{(b)} - z_{\text{tgt}}^{(b)}\big\|_2^2 + \lambda_{\text{ent}}\mathcal{R}_{\text{spectral}}(\mathbf{K}) + \beta_{\text{Lyap}}\mathcal{L}_{\text{Lyap}}, \quad (3)$$

where B is the batch size, $\beta_{\text{norm}} = 1.0$, $\lambda_{\text{ent}} = 0.5$, and $\beta_{\text{Lyap}} = 0.02$ weights the discrete Lyapunov defect $\mathcal{L}_{\text{Lyap}} = D^{-1}\|\mathbf{K}\tilde{\mathbf{C}}_{\text{ctx}}\mathbf{K}^\top - \mathbf{C}_{\text{tgt}}\|_F^2$ on trace-normalized batch covariances $\tilde{\mathbf{C}}_{\text{ctx}} = \text{sg}(\text{Cov}(z_{\text{ctx}}))/(\text{tr}(\text{Cov}(z_{\text{ctx}}))/D + 10^{-6})$ and $\mathbf{C}_{\text{tgt}} = \text{Cov}(z_{\text{tgt}})/(\text{tr}(\text{Cov}(z_{\text{tgt}}))/D + 10^{-6})$ using stop-gradient $\text{sg}(\cdot)$. We verify both finite-ϵ repulsion and non-degeneracy across the complete objective.

Proposition 1 (Operator Non-Degeneracy, Lyapunov Invariance, and Finite-ϵ Energy Bounds)**.** *Let* $\mathbf{M} = (BD)^{-1}\sum_{b=1}^{B} z_{tgt}^{(b)} z_{ctx}^{(b)\top}$ *denote the cross-covariance under (3) with* $\lambda_{ent}, \beta_{norm} > 0$ *and* $\epsilon \in (0,1)$*: (i)* Vanishing Lyapunov origin gradient and non-stationarity for $\mathbf{M} \neq \mathbf{0}$: *At* $\mathbf{K} = \mathbf{0}$, $\nabla_{\mathbf{K}}\mathcal{R}_{spectral}(\mathbf{0}) = \mathbf{0}$, *and because* $\mathbf{K}\tilde{\mathbf{C}}_{ctx}\mathbf{K}^\top$ *is quadratic in* $\mathbf{K}$, *the derivative* $\nabla_{\mathbf{K}}\mathcal{L}_{Lyap}(\mathbf{0}) = \mathbf{0}$ *vanishes identically. Thus* $\nabla_{\mathbf{K}}\mathcal{L}_{Op\text{-}Entropy}(\mathbf{0}) = -2\mathbf{M} \neq \mathbf{0}$, *so* $\mathbf{K} = \mathbf{0}$ *is not stationary under the full objective. (ii)* Strict local maximum when $\mathbf{M} = \mathbf{0}$: *Even if* $\mathbf{M} = \mathbf{0}$, *for any* $\mathbf{K}$ *with* $r(\mathbf{K}) < 1 - \epsilon$, $\nabla_{\mathbf{K}}\Phi_{energy}(\mathbf{K}) = -\frac{2(1-r(\mathbf{K})-\epsilon)}{D(r(\mathbf{K})+\epsilon)}\mathbf{K}$ *points strictly radially outward, and the origin Hessian* $\nabla^2_{\mathbf{K}}\Phi_{energy}(\mathbf{0}) = -\frac{2(1-\epsilon)}{D\epsilon}\mathbf{I}_{D^2} \prec \mathbf{0}$ *is strictly negative definite, making* $\mathbf{K} = \mathbf{0}$ *an unstable repeller. (iii)* Finite-ϵ energy lower bound and asymptotic barrier: *Over* $r \geq 0$, Φ_{energy} *attains its global minimum at* $r^* = 1 - \epsilon$ *with value* $\Phi_{energy}(r^*) = \ln(1+\epsilon) - \epsilon = -\frac{\epsilon^2}{2} + \mathcal{O}(\epsilon^3) \approx -5 \times 10^{-13}$ *for* $\epsilon = 10^{-6}$ *(non-negative under standard numerical clamping* $\max(\Phi_{energy}, 0)$*). At* $\mathbf{K} = \mathbf{0}$, *the finite barrier is* $\Phi_{energy}(\mathbf{0}) = \ln(1+\epsilon) - 1 - \ln\epsilon \approx 12.82$ *nats with steep restoring slope* $\partial\Phi_{energy}/\partial r|_{r=0^+} = 1 - 1/\epsilon \approx -10^6$, *recovering an infinite barrier in the asymptotic limit* $\lim_{\epsilon\to 0^+}\Phi_{energy}(\mathbf{0}) = +\infty$.

Proof. At $\mathbf{K} = \mathbf{0}$, spectral probabilities $p_i = 1/D$ yield $\nabla_{\mathbf{K}}\Delta\mathcal{H}(\mathbf{0}) = \mathbf{0}$. For $\mathcal{L}_{\text{Lyap}}$, let $\mathbf{\Delta}(\mathbf{K}) = \mathbf{K}\tilde{\mathbf{C}}_{\text{ctx}}\mathbf{K}^\top - \mathbf{C}_{\text{tgt}}$. Expanding the directional derivative in direction $\mathbf{V}$ gives $\langle\nabla_{\mathbf{K}}\mathcal{L}_{\text{Lyap}}(\mathbf{K}), \mathbf{V}\rangle = \frac{4}{D}\,\text{tr}(\mathbf{\Delta}(\mathbf{K})\tilde{\mathbf{C}}_{\text{ctx}}\mathbf{V}^\top)$, which vanishes at $\mathbf{K} = \mathbf{0}$, confirming $\nabla_{\mathbf{K}}\mathcal{L}_{\text{Lyap}}(\mathbf{0}) = \mathbf{0}$. By the chain rule on (2), $\nabla_{\mathbf{K}}\Phi_{\text{energy}}(\mathbf{K}) = -\frac{2(1-r(\mathbf{K})-\epsilon)}{D(r(\mathbf{K})+\epsilon)}\mathbf{K}$, vanishing at $\mathbf{K} = \mathbf{0}$ (leaving $-2\mathbf{M}$ when $\mathbf{M} \neq \mathbf{0}$) with negative-definite Jacobian $-2(1-\epsilon)(D\epsilon)^{-1}\mathbf{I}_{D^2} \prec \mathbf{0}$. Differentiating $\Phi_{\text{energy}}(r)$ gives $d\Phi_{\text{energy}}/dr = 1 - (r+\epsilon)^{-1}$, vanishing at $r^* = 1-\epsilon$ where $\Phi_{\text{energy}}(1-\epsilon) = \ln(1+\epsilon) - \epsilon \approx -5\times 10^{-13}$. Evaluating at $r = 0, \epsilon = 10^{-6}$ yields 12.82 nats, with $\lim_{\epsilon\to 0^+}[-\ln\epsilon] = +\infty$. □

Remark (Encoder Non-Collapse): While Proposition 1 prevents collapse of $\mathbf{K}$, encoder collapse ($z_{\text{ctx}}, z_{\text{tgt}} \to \mathbf{c}$) is

prevented structurally by terminal LayerNorm ($\|z\|_2 = \sqrt{D}$), EMA momentum with stop-gradient decoupling, and data-projected covariance entropy on $\mathbf{K}z_{\text{ctx}}$ ($\mathrm{RankMe}(z_{\text{tgt}}) \approx 11.6$, Fig. 3(a)).

2) Reynolds-Stress JEPA (Second-Moment Subgrid Closure): **Physical Motivation and Structural Parsimony.** Inspired by Reynolds averaging in turbulence, which splits a flow into its resolved mean $\bar{z}$ and subgrid covariance $\mathbf{\Sigma}$, Reynolds-Stress JEPA predicts both moments while enforcing stability and positive semidefinite (PSD) geometry directly inside the network layers:

1) *Cayley-accretive Crouzeix linear core and stability bounds:* The predictor separates linear transport from nonlinear shear: $\hat{z}_{\text{tgt}} = \mathbf{W}_{\text{Crouzeix}} z_{\text{ctx}} + \tanh(\gamma_{\text{shear}})\psi_{\text{MLP}}(z_{\text{ctx}})$. The linear core parameterizes a strictly accretive generator $\mathbf{G} = \mathbf{S}_{\text{skew}} + \mathbf{D}_{\text{visc}}$, with skew-symmetric $\mathbf{S}_{\text{skew}} = 0.5(\mathbf{A} - \mathbf{A}^\top)$ and PSD $\mathbf{D}_{\text{visc}} = D^{-1}\mathbf{B}\mathbf{B}^\top + \nu_0\mathbf{I}_D \succeq \nu_0\mathbf{I}_D$ ($\nu_0 = 10^{-3}$), and applies the Cayley transform $\mathbf{W}_{\text{Crouzeix}} = (\mathbf{I} + \mathbf{G})^{-1}(\mathbf{I} - \mathbf{G})$. Because $\mathrm{Re}\langle u, \mathbf{G}u\rangle = \langle u, \mathbf{D}_{\text{visc}}u\rangle \geq \nu_0\|u\|_2^2$, substituting $u = (\mathbf{I} + \mathbf{G})^{-1}v$ into $\|(\mathbf{I} + \mathbf{G})u\|_2^2 - \|(\mathbf{I} - \mathbf{G})u\|_2^2 = 4\,\mathrm{Re}\langle u, \mathbf{G}u\rangle \geq 4\nu_0\|u\|_2^2$ proves that the linear core is a strict Euclidean operator-norm contraction: $\|\mathbf{W}_{\text{Crouzeix}}\|_2 \leq (1 - \nu_0)/(1 + \nu_0) < 1$ (0.9980). This places its numerical range $W(\mathbf{W}_{\text{Crouzeix}}) \subset \mathbb{D}$ strictly inside the open unit disk. By Crouzeix's theorem [38], $\|p(\mathbf{W}_{\text{Crouzeix}})\|_2 \leq (1 + \sqrt{2})\sup_{z\in\mathbb{D}}|p(z)|$ for every polynomial p. Across trained models, numerical verification confirms $\|\mathbf{W}_{\text{Crouzeix}}\|_2 \in [0.998006, 0.998009]$, spectral radius $\rho(\mathbf{W}_{\text{Crouzeix}}) \in [0.996449, 0.997700]$, and numerical radius $r_{\text{num}}(\mathbf{W}_{\text{Crouzeix}}) \in [0.997639, 0.997841]$, while nonlinear shear corrections remain perturbative ($|\tanh(\gamma_{\text{shear}})| \in [0.014, 0.033]$).
2) *Sign-stabilized QR Stiefel retraction and PSD cone closure:* Alongside $\bar{z}_{\text{tgt}} \in \mathbb{R}^D$, the encoder centers the 64-channel TCN sequence $h(\tau) - \bar{h}$ and projects it via $\mathbf{Q}_{\text{St}} \in \mathrm{St}(32, 64)$ into fluctuations $z'_{\text{tgt}}(\tau) \in \mathbb{R}^D$, which a subgrid frame $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_{d_s}]^\top \in \mathrm{St}(d_s, 32)$ ($d_s = 16$) compresses into $\mathbf{\Sigma}_{\text{obs}} \in \mathcal{S}_+^{d_s}$. Both $\mathbf{Q}_{\text{St}}$ and $\mathbf{V}$ are initialized to the truncated identity and retracted onto the column-orthonormal Stiefel manifold via sign-stabilized reduced QR factorization ($\mathbf{W}^\top = \mathbf{Q}\mathbf{R}$, multiplying column i of $\mathbf{Q}$ by $\mathrm{sign}(R_{ii})$ with $0 \mapsto +1$ to eliminate QR sign-flip gradient discontinuities). On $\bar{z} = 0.5(z_{\text{ctx}} + \hat{z}_{\text{tgt}})$, the closure head predicts $\mathbf{\Sigma}_{\text{pred}} = \sum_{k=1}^{d_s} c_k(\bar{z})\mathbf{v}_k\mathbf{v}_k^\top + \gamma\mathbf{I}_{d_s} \in \mathcal{S}_{++}^{d_s}$ with positive weights $c_k = \mathrm{softplus}(w_k(\bar{z})) + 10^{-4}$ and isotropic floor $\gamma = 10^{-3}$ ($\lambda_{\min}(\mathbf{\Sigma}_{\text{pred}}) \geq 10^{-3}$ strictly enforced by construction) under the two-term objective:

$$\begin{aligned}\mathcal{L}_{\text{Reynolds}} = &\frac{1}{BD}\sum_{b=1}^{B}\big\|\hat{z}_{\text{tgt}}^{(b)} - z_{\text{tgt}}^{(b)}\big\|_2^2 \\ &+ \frac{\alpha_{\text{stress}}}{Bd_s^2}\sum_{b=1}^{B}\big\|\mathbf{\Sigma}_{\text{pred}}^{(b)} - \mathrm{sg}\big(\mathbf{\Sigma}_{\text{obs}}^{(b)}\big)\big\|_F^2,\end{aligned} \tag{4}$$

with $\alpha_{\text{stress}} = 0.1$ and no auxiliary loss penalties.

C. Anomaly Scoring and Operational Detection Latency

At test time, windows are scored at stride 1. Let $\mathbf{e}(t) = z_{\text{tgt}}(t) - \hat{z}_{\text{tgt}}(t) \in \mathbb{R}^D$ denote the latent residual and $g(\mathbf{e}) \equiv g_D(\mathbf{e})$ the dimension-agnostic coordinate saliency gate defined in (7). Unconstrained `TS-JEPA` uses $\mathcal{S}(t) = g(\mathbf{e}(t))$. Because terminal LayerNorm strips radial amplitude, `Operator-Entropy JEPA` tracks pre-LayerNorm energy ($e_{\text{ctx}}, e_{\text{tgt}}$) with $e_{\text{pred}} = e_{\text{ctx}}\|\hat{z}_{\text{tgt}}\|_2/(\|z_{\text{ctx}}\|_2 + 10^{-6})$, while `Reynolds-Stress JEPA` adds 4-octave dyadic shell-energy leakage $\Delta E_{\text{LP}}(t)$ and saliency-gated stress discrepancy ($k_s = \min(K, d_s)$):

$$\mathcal{S}_{\text{OpEnt}}(t) = g\big(\mathbf{e}(t)\big) + 0.2\,\frac{|e_{\text{tgt}}(t) - e_{\text{pred}}(t)|}{e_{\text{tgt}}(t) + e_{\text{pred}}(t) + 10^{-6}}, \tag{5}$$

$$\begin{aligned}\mathcal{S}_{\text{Rey}}(t) = &\,g\big(\mathbf{e}(t)\big) + 0.2\,\Delta E_{\text{LP}}(t) \\ &+ \frac{0.2}{\sqrt{k_s}}\,g_{d_s^2}\big(\mathrm{vec}(\mathbf{\Sigma}_{\text{obs}} - \mathbf{\Sigma}_{\text{pred}})\big),\end{aligned} \tag{6}$$

where $g_{d_s^2}$ operates on the vectorized $d_s \times d_s$ covariance departure ($d_s^2 = 256$) normalized by $\sqrt{k_s}$ to preserve scale compatibility with g_D.

Three-Part Operational Causality Framework: We formalize online evaluation under three criteria (Fig. 2):

(a) **Data causality ($\mathcal{F}_\tau$-measurability):** At decision timestamp $\tau = t + S - 1$, the detector observes only $\mathbf{X}[t - C : \tau, :]$; neither encoder accesses observations beyond τ.

(b) **Score-emission latency ($S - 1 = 63$ steps) vs. fault-detection delay (Δt):** Because z_{tgt} summarizes suspect observations $\mathbf{X}[t : t + S, :]$ spanning sample indices $[t, t + S - 1]$, score s_t is emitted at the trailing edge $\tau = t + S - 1$ (Fig. 2, Row 3), creating a structural emission lag of $S - 1 = 63$ steps (0.98 s on 64 Hz Daphnet, 10.5 min on 0.1 Hz SMAP/MSL/GHL, 63 min on GECCO, and 63 h on hourly Metro). This lag is distinct from empirical fault-detection delay $\Delta t = t_{\text{alarm}} - t^*$ from onset t^* to the first alarm $t_{\text{alarm}} \geq t^*$. Shortening the horizon to $S \in \{1, 4\}$ eliminates emission latency for rapid tripwires (Section VII-D).

(c) **Strictly causal trailing vs. retrospective buffered attribution:** *Causal trailing attribution* assigns s_t only to $\tau = t + S - 1$ for zero-lookahead streaming. *Retrospective buffered smear attribution* averages s_t across all suspect timestamps $[t, t + S - 1]$, introducing fixed-lag smoothing. Both regimes are reported separately throughout (Table III(a),(b)).

D. Shared Architectural Components and Thresholding

1) Coordinate Saliency Gating and Scale Equivariance: To amplify sparse departures in high-dimensional representations while mitigating channel drowning, we define the coordinate saliency gate $g_d : \mathbb{R}^d \to \mathbb{R}_{\geq 0}$ for any vector $\mathbf{v} \in \mathbb{R}^d$:

$$\begin{aligned}g_d(\mathbf{v}) &= (1 - \alpha)\|\mathbf{v}\|_2 + \alpha\sqrt{d\sum_{i=1}^{d} w_i v_i^2}, \\ w_i &= \frac{\exp\big(|v_i|/(\tau_g\sigma_{\mathbf{v}})\big)}{\sum_{j=1}^{d}\exp\big(|v_j|/(\tau_g\sigma_{\mathbf{v}})\big)}, \quad \sigma_{\mathbf{v}} = \frac{\|\mathbf{v}\|_2}{\sqrt{d}} + \epsilon,\end{aligned} \tag{7}$$

with $\alpha = 0.5$, $\tau_g = 0.5$, and $\epsilon = 10^{-6}$. When $\mathbf{v} = \mathbf{0}$, $g_d(\mathbf{0}) = 0$. For non-zero $\mathbf{v}$, rescaling $\mathbf{v} \mapsto c\mathbf{v}$ with $c > 0$ yields $\sigma_{c\mathbf{v}} = c\frac{\|\mathbf{v}\|_2}{\sqrt{d}} + \epsilon = c\sigma_{\mathbf{v}} + (1 - c)\epsilon$. In the operational regime where $\|\mathbf{v}\|_2 \gg \epsilon$, the softmax weights w_i are scale-invariant up to

an $\mathcal{O}(\epsilon/\|\mathbf{v}\|_2)$ perturbation, satisfying approximate degree-1 homogeneity $g_d(c\mathbf{v}) \approx c\, g_d(\mathbf{v})$ and recovering exact degree-1 homogeneity in the limit $\epsilon \to 0^+$. Input channels use uniform weights in the deployed pipeline (Section VI-C).

2) Multiscale Frequency Front-End: For multivariate streams ($K > 1$), a causal depthwise 1D convolutional stem filters each channel prior to 1×1 mixing ($K \leq 1$ uses an identity bypass). Each projected channel is split across $J = 4$ smooth dyadic shells on a 256-point rFFT grid normalized into an exact amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ within 1.2×10^{-7}), processed by causal 1D convolutions (kernels $\{15, 9, 5, 3\}$), and recombined via a softmax scale router. Because smooth Fourier shells overlap at band edges ($\sum \Psi_j \equiv 1$ rather than $\sum \Psi_j^2 \equiv 1$), this acts as a multiscale front-end and frequency-domain leakage feature (ΔE_{LP}) rather than an orthogonal Parseval wavelet bank (Table II(b)).

3) Extreme-Value Threshold Calibration (SPOT): To convert scores $\mathcal{S}(t)$ into binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_{\text{SPOT}})$ from nominal calibration data alone, Streaming Peaks-Over-Threshold (SPOT) [7] fits exceedances $Y = \mathcal{S}(t) - u > 0$ above the 98th calibration percentile $u = p_{98}$ to a Generalized Pareto Distribution (GPD) with shape ξ and scale σ via Grimshaw maximum-likelihood estimation [39] with bounded L-BFGS-B fallback ($\xi \in [-0.9, 10]$). Inverting the GPD tail at target risk $q = 10^{-3}$ yields:

$$\tau_{\text{SPOT}} = \begin{cases} u + \frac{\sigma}{\xi}\left[\left(\frac{qN_{\text{cal}}}{N_u}\right)^{-\xi} - 1\right], & |\xi| \geq 10^{-5}, \\ u - \sigma \ln\left(\frac{qN_{\text{cal}}}{N_u}\right), & |\xi| < 10^{-5}, \end{cases} \tag{8}$$

where the lower branch is the continuous exponential-tail limit as $\xi \to 0$, evaluated directly to avoid division by zero. Here N_{cal} and N_u count post-attribution score timestamps on the calibration block. Primary tables use standard SPOT; Section VII-F evaluates runs-declustered GPD fitting, rolling calibrators, and kurtosis-scaled thresholds (τ_{eff} and τ_{exc}).

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup and Hyperparameters

All architectures use an identical streaming pipeline with context window $C = 256$ and target horizon $S = 64$ ($W = C + S = 320$), trained at stride 16 and evaluated at stride 1. Context and target representations ($D = 32$) are extracted by the multiscale frequency front-end followed by a 6-block causal dilated TCN (64 filters, kernel size 5, dilations $\{1, 2, 4, 8, 16, 32\}$). Multi-statistic temporal pooling (terminal, mean, max, standard deviation, Log-Ricci energy, and first-order total variation) projects features through a linear bottleneck before LayerNorm and dropout ($p = 0.20$). Models are trained for 20 epochs on nominal sequences using AdamW (learning rate $10^{-3} \to 10^{-5}$ via cosine annealing, weight decay 10^{-4}, batch size 32), with identical training budgets and zero per-dataset hyperparameter tuning across families.

a) Held-Out Calibration Protocol and Boundary Isolation: To evaluate out-of-sample threshold calibration without circularity, each stream's raw nominal sequence $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Row 1) into

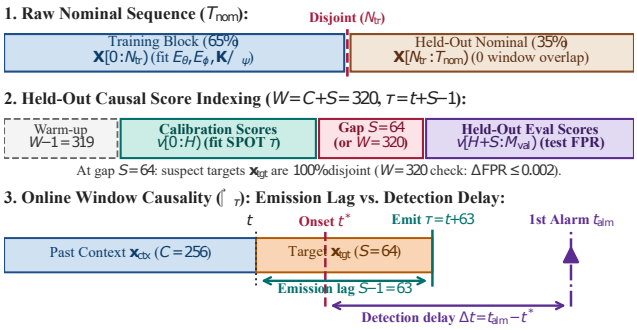


Fig. 2. Chronological split, boundary window indexing, and online causality protocol (C=256, S=64, W=320). Row 1 separates raw training (65%) and held-out nominal (35%) observations with zero window overlap across N_{tr}. Row 2 indexes causal held-out scores $v[0 : M_{\text{val}})$ into context warm-up ($W-1$=319), calibration $v[0 : H)$, isolation gap S=64 (or full-window W=320), and held-out evaluation $v[H+S : M_{\text{val}})$. Row 3 distinguishes structural score-emission lag ($S-1$=63) from empirical fault-detection delay ($\Delta t = t_{\text{alm}} - t^*$).

a *training block* $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = \lfloor 0.65\, T_{\text{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$. Standardization parameters and neural weights are fit on $\mathbf{X}_{\text{tr}}$ windows alone, guaranteeing zero raw-observation or window overlap across N_{tr}.

Within $\mathbf{X}_{\text{val}}$ ($N_{\text{val}} = T_{\text{nom}} - N_{\text{tr}}$), sliding windows of length $W = 320$ are constructed strictly inside $\mathbf{X}_{\text{val}}$ (Fig. 2, Row 2). Under causal trailing attribution, the first $W-1 = 319$ observations serve as context warm-up, yielding $M_{\text{val}} = N_{\text{val}} - W + 1$ valid score timestamps $v[0 : M_{\text{val}})$. Splitting v at midpoint $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the *nominal calibration block* $v_{\text{cal}} = v[0 : H)$ (used to fit the SPOT threshold at $u = p_{98}, q = 10^{-3}$) and the *held-out evaluation block* $v_{\text{hold}} = v[H + S : M_{\text{val}})$, excluding $S = 64$ intermediate windows $v[H : H + S)$. Because suspect target windows have length $S = 64$, the last calibration target and first evaluation target are 100% disjoint; the first $C = 256$ evaluation steps share passive historical context with calibration observations, mirroring online streaming where past nominal history forms context for current predictions. A stricter full-window isolation check ($G = W = 320$, excluding $v[H : H + 320)$ so even passive context is disjoint across the seven streams with $M_{\text{val}}/2 > 420$) leaves stream-macro causal FPR virtually unchanged (2.07% $\to$ 2.18% for OpEntropy and 3.02% $\to$ 3.24% for TS-JEPA versus 12.15% $\to$ 13.97% for TimesNet and 9.05% $\to$ 9.73% for TranAD). All primary tables use fixed standard SPOT thresholds during testing.

b) Independent Statistical Units, Multi-Seed Replication, and Equivalence Bounds: Principal models on the 19 non-GHL streams are evaluated across five random seeds ($\{42, 123, 456, 7, 2024\}$); the 14 GHL streams ($> 10^6$ observations each) and ablation controls use three seeds ($\{7, 42, 2024\}$). To prevent pseudoreplication, every statistical test specifies its independent unit explicitly: (i) seeds are averaged within each stream first to quantify within-stream optimization variance (SD_{seed}), (ii) stream-level paired tests treat the telemetry stream ($n = 33$ primary or $n = 19$ non-GHL) as the unit of analysis, and (iii) domain-level tests aggregate streams within each hardware platform first and treat the dataset family ($N = 7$) as the independent cluster. We employ three statistical tools:

1) **Paired Wilcoxon signed-rank test and false-discovery-**

rate adjustment: The Wilcoxon signed-rank test evaluates whether the median paired difference across streams ($n = 33$ or $n = 19$) is zero without Gaussian assumptions. We adjust p-values across multiple comparisons via the Benjamini–Hochberg false-discovery-rate (FDR) procedure ($q_{\text{FDR}} = 0.05$), report 95% stream-resampled bootstrap confidence intervals (CIs; 2,000 resamples), and cross-check at the dataset-cluster level ($N = 7$) via exact sign-flip permutation tests and zero-tuning LODO sensitivity.

2) **Two One-Sided Tests (TOST) for statistical equivalence:** To test whether a physics-inspired regularizer produces practically negligible change relative to an unregularized control on the same backbone ($n = 19$ streams), TOST tests $H_{01} : \mu_1 - \mu_2 \leq -\Delta$ (p_{lower}) and $H_{02} : \mu_1 - \mu_2 \geq +\Delta$ (p_{upper}), establishing equivalence when $p_{\text{TOST}} = \max(p_{\text{lower}}, p_{\text{upper}}) < 0.05$ (i.e., the 90% paired CI lies strictly inside $[-\Delta, +\Delta]$). We prespecify $\Delta = \pm 0.030$ Point-F1 ($\approx 13\%$ of baseline F1, $< 0.09\times$ between-stream SD) and report sensitivity across $\Delta \in \{\pm 0.010, \pm 0.015, \pm 0.020\}$, confirming equivalence down to $\Delta = \pm 0.015$ (Table II(a)).

3) **Distributional tail and drift diagnostics (γ_2 and D_{KS}):** On nominal windows we measure: (i) *excess kurtosis* $\gamma_2 = \mathbb{E}[(s - \mu)^4]/\sigma^4 - 3$, distinguishing heavy-tailed *leptokurtic* ($\gamma_2 > 0$) from light-tailed *platykurtic* ($\gamma_2 < 0$) residuals; and (ii) the two-sample *Kolmogorov–Smirnov (KS) drift statistic* $D_{\text{KS}} = \sup_x |F_{\text{cal}}(x) - F_{\text{hold}}(x)| \in [0, 1]$, quantifying distributional shift between calibration (F_{cal}) and held-out nominal (F_{hold}) blocks.

B. Baseline Architectures, Fairness Checks, and Cohort Structure

We separate primary multi-seed conclusions (Cohorts 1–2 across 14 neural, linear, and non-parametric arms) from secondary single-seed cross-checks. Two fairness audits verify that reconstructive baseline results under the unified 20-epoch, $W = 320$ pipeline are not artifacts of window length or implementation:

1) **Native published window lengths** (5 streams): Evaluating TranAD ($W = 10$) and TimesNet ($W = 96$) at their published window sizes raises Point-F1 (0.164 and 0.218), but observation residuals remain severely leptokurtic ($\gamma_2 = 448.3$ and 98.1 vs. 3.1 for TS-JEPA) with high nominal FPR (28.4% and 44.1%).
2) **Secondary 130-stream single-seed cross-check:** Reconstructors replicate their published point-adjusted F1 on SMD and MSL, and Table III(d) provides a secondary 130-stream, 11-dataset single-seed comparison against NCAD, DCdetector, and Anomaly Transformer.

C. Cohort Summary and Reproducibility Specifications

Experiments are organized into five cohorts with distinct statistical roles:

- **Cohort 1 (33-stream primary multi-seed benchmark):** 19 non-GHL streams (5 seeds) and 14 GHL streams (3 seeds), reported in Tables III(a)–(c) and IV(a).
- **Cohort 2 (19-stream matched-backbone factorial and operating-curve suite):** 14 controlled arms across 5 seeds, reported in Table II(a),(c) and Fig. 4(b).
- **Cohort 3 (6-stream operator sweep, quiet-fault, and hybrid-monitor suite):** Factorial regularizer sweep (3 seeds) and six injected fault types with hybrid sensor monitors, reported in Fig. 3(a),(b).
- **Cohort 4 (5-stream horizon-latency and frequency-shell panels):** Target-horizon sweep ($S \in \{1, 4, 16, 32, 64\}$, 3 seeds, matched emission lag $S-1$) and dyadic front-end ablation, reported in Fig. 4(a) and Table II(b).
- **Cohort 5 (130-stream single-seed baseline screen):** Secondary breadth cross-check across 11 datasets against NCAD, DCdetector, and Anomaly Transformer (Table III(d)), treated strictly as auxiliary evidence rather than equivalent to the multi-seed benchmark.

At $K = 25$ channels, online trainable parameter counts are closely matched across latent and observation-space backbones: 272,266 online parameters for TS-JEPA (540,148 total including the stop-gradient EMA target encoder), 268,906 online (536,788 total) for OpEntropy JEPA, 281,732 online (549,614 total) for Reynolds-Stress JEPA, 304,138 for matched-encoder `TCN-ObsPred`, 307,402 for `TCN-ObsPred-LN`, 648,553 for TimesNet, and 205,746 for TranAD. Non-finite telemetry entries are zero-imputed with channel standard deviations floored at 10^{-2}. Causal and retrospective buffered scores receive separately fitted thresholds τ_{SPOT} fitted strictly within their respective calibration windows v_{cal}, where N_{cal} and N_u count post-attribution score timestamps rather than raw telemetry samples. GPD parameters (ξ, σ) above $u = p_{98}$ are estimated via Grimshaw maximum-likelihood estimation (MLE) with a bounded quasi-Newton (L-BFGS-B) fallback ($\xi \in [-0.9, 10], \sigma \geq 10^{-6}$), switching continuously to the exponential tail limit for $|\xi| < 10^{-5}$. All 195 primary multi-seed runs converged under the 20-epoch schedule on PyTorch 2.4 / CUDA 12.4 (NVIDIA RTX 5090, 0.60–0.84 ms per window, 184 MB of GPU memory); stream manifests, fitted thresholds, and source code are available at https://github.com/quaid281/NCAD.

D. Multi-Tier Statistical Evaluation Metrics

To guard against point-adjustment inflation, channel-averaging bias, and threshold gaming [10], [31], we evaluate models across three tiers: (i) *primary calibration and accuracy outcomes*, comprising realized held-out nominal FPR (FPR_{hold}), out-of-sample calibration error $|\text{FPR}_{\text{hold}} - q|$ relative to design risk $q = 10^{-3}$, unadjusted Point-F1, and threshold-free PR-AUC; (ii) *operating characteristics*, evaluating event recall across deployable calibration quantiles (q^*, Fig. 4(b)) and across post-hoc matched realized test-FPR budgets ($f^* \in \{0.01, 0.05\}$, Table II(c)); and (iii) *secondary operational endpoints*, reporting unweighted Dataset-Macro Point-F1, event recall (fraction of contiguous anomaly segments with ≥ 1 alarm), mean detection delay Δt over detected events, and the constant-alarm baseline $\text{F1}_{\text{all-pos}} = 2\pi/(1 + \pi)$ at prevalence π.

TABLE II

ABLATION AND MECHANISM-DISCRIMINATION STUDIES: (A) FIXED-BACKBONE REGULARIZER CONTROL AND TOST EQUIVALENCE (n=19, 5 SEEDS), (B) DYADIC FRONT-END (n=5), AND (C) 14-ARM MATCHED-BACKBONE FACTORIAL COMPARISON (n=19, 5 SEEDS). ER IS EVENT RECALL.

(a) Regularizer Control on Fixed Architecture & TOST Equivalence

Model / Control	Pt-F1	PR	FPR	ER	ΔF1 [90%]	p_{TOST}
TS-JEPA (Unconstrained)	0.206 ± 0.025	0.331	0.093	0.477	Baseline	–
Operator-Entropy JEPA	0.233 ± 0.028	0.351	0.101	0.550	-0.004 $[-.011, +.003]$	0.0007*
$\lambda_{\text{ent}} = 0$ control	0.235 ± 0.038	0.365	0.099	0.580	–	($0.0024^{\dagger}$)
Reynolds-Stress JEPA	0.148 ± 0.022	0.285	0.190	0.421	-0.009 $[-.052, +.032]$	$0.211^{\ddagger}$
$\alpha_{\text{stress}} = 0$ control	0.167 ± 0.012	0.281	0.109	0.419	–	–
TimesNet (Baseline)	0.197 ± 0.018	0.329	0.300	0.583	–	–
TranAD (Baseline)	0.130 ± 0.004	0.287	0.307	0.490	–	–

*TOST equiv. at $\Delta = \pm 0.030$ ($p<0.001$); †equiv. to $\Delta = \pm 0.015$ ($p=0.0024$; at ± 0.010, $p=0.086$); ‡TOST $p=0.211$ at $\Delta=\pm 0.030$ (3 matched seeds; two-sided Wilcoxon $p=0.758$). Matched 3-seed Reynolds mean is 0.158 (ΔF1$=-0.009$).

(b) Dyadic-Shell Front-End On/Off

Stream (F1 / PR)	Std. Conv	Dyadic	ΔF1 / ΔPR
SMAP `P-3`	**0.301** / 0.280	0.115 / **0.296**	-0.186 / $+0.017$
MSL `M-1`	0.014 / **0.491**	**0.016** / 0.449	$+0.002$ / -0.043
SMD `machine-1-2`	0.000 / 0.028	**0.068** / **0.221**	$+0.068$ / $+0.193$
Daphnet `S01R01E1`	**0.179** / **0.103**	0.130 / 0.066	-0.048 / -0.037
GECCO `water_quality`	**0.097** / 0.161	0.062 / **0.282**	-0.035 / $+0.121$
Macro Mean	**0.118** / 0.213	0.078 / **0.263**	-0.040 / $+0.050$

(c) Controlled 14-Arm Mechanism-Discrimination Study Across 19 Non-GHL Streams (5 Seeds)

	Tail / Drift		Held-Out FPR		Point-F1		PR-AUC		Causal Event Recall			
Architecture Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	SPOT	q^*=0.05	f^*=0.01	f^*=0.05
`TS-JEPA` (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
`OpEntropy JEPA`	0.94	0.209	0.028	0.101	0.153	0.233	0.280	**0.351**	0.517	0.752	0.371	0.573
`Reynolds-Stress JEPA`	2.04	0.274	0.086	0.190	0.060	0.148	0.231	0.285	0.474	0.667	0.281	0.494
`Reynolds` (α_{stress}=0)	2.41	0.193	0.019	0.109	0.051	0.167	0.220	0.281	0.424	0.756	**0.457**	**0.633**
`JEPA-CtxRecon` (Context Autoenc.)	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
`JEPA-NoEMA` (Hard-Copy Target)	1.16	**0.167**	**0.009**	0.098	0.146	**0.235**	0.273	0.328	0.503	**0.816**	0.398	0.600
`TS-JEPA-NoGate` (Plain ℓ_2 Norm)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
`PCA-LinPred-LN` (Linear Subspace)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
`TCN-ObsPred-LN` (Obs. + LayerNorm)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
`IsoForest` (Non-Parametric)	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
`TCN-ObsPred` (Unbounded Obs. Pred.)	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
`TCN-ObsRecon` (Unbounded Obs. Recon.)	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
`TimesNet` (Baseline)	5.54	0.438	0.243	0.300	**0.209**	0.197	**0.293**	0.329	**0.588**	0.618	0.217	0.407
`TranAD` (Baseline)	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

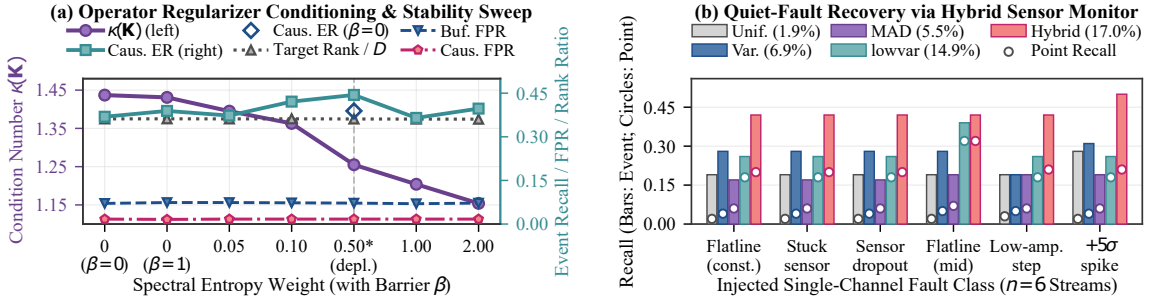


Fig. 3. Operator regularizer conditioning sweep and quiet-fault recovery (n=6 streams, three seeds). **(a)** Condition number $\kappa(\mathbf{K})$, target effective rank ratio $\text{RankMe}(z_{\text{tgt}})/D$, event recall (ER), and false-positive rate (FPR) across $\lambda_{\text{ent}} \in [0, 2.0]$ with Bregman barrier $\beta \in \{0, 1\}$. **(b)** Injected single-channel event recall (bars) and point recall (circles) across uniform TS-JEPA (`Unif.`), context variance (`Var.`) and median absolute deviation (`MAD`) gates, standalone rolling variance monitor (`lowvar`), and the hybrid union monitor (`TS-JEPA` $\vee$ `lowvar`; nominal FPR in parentheses).

VI. ABLATION STUDIES AND ANALYSIS

A. Ablation 1: Operator Regularizer Control on Fixed Architecture

To test whether the spectral regularizer drives anomaly detection accuracy on its fixed Koopman backbone, Table II(a) compares Operator-Entropy JEPA against an unregularized control ($\lambda_{\text{ent}} = 0$) across all 19 non-GHL streams. Removing the spectral regularizer leaves buffered Point-F1 (0.233 vs. 0.235, paired difference -0.004, two-sided $p = 0.89$) and PR-AUC (0.351 vs. 0.365) virtually identical. Two One-Sided Tests confirm formal equivalence within the prespecified $\Delta = \pm 0.030$ margin ($p_{\text{TOST}} = 0.0007$) and across sensitivity checks down to $\Delta = \pm 0.020$ ($p_{\text{TOST}} = 0.0015$) and $\Delta = \pm 0.015$ ($p_{\text{TOST}} = 0.0024$), where the 90% CI $[-0.0112, +0.0029]$ lies strictly inside $[-\Delta, +\Delta]$. At the tighter $\Delta = \pm 0.010$ margin, the parametric lower bound extends marginally outside the bound ($p_{\text{TOST}} = 0.086$), while equivalence holds within ± 0.010 PR-AUC and ± 0.025 event recall.

This fixed-backbone regularizer control aligns with findings on the Reynolds-Stress JEPA architecture: isolating the base predictive residual from the quadratic stress closure discrepancy ($\alpha_{\text{stress}} = 0$) leaves buffered PR-AUC (0.281 vs. 0.285) and Point-F1 (0.167 vs. 0.148 across all seeds; 0.167 vs. 0.158 on matched seeds, paired difference -0.009, 90% CI $[-0.052, +0.032]$, TOST $p = 0.211$, two-sided Wilcoxon $p = 0.758$) comparable to the full Reynolds model while achieving 0.019 causal held-out FPR and leading matched-FPR event recall (0.457 at f^*=0.01 and 0.633 at f^*=0.05, Table II(a),(c)). Across both physics-inspired architectures, regularizer penalties condition latent representations without driving aggregate detection gains, confirming that false-alarm suppression is governed by the joint-embedding prediction framework rather than penalty additions.

B. Ablation 2: Factorial Isolation of the Entropy Term and Energy Barrier

Fig. 3(a) crosses the spectral-entropy term and Bregman energy barrier across $\lambda_{\text{ent}} \in [0, 2.0]$ on six streams (three seeds), tracking effective rank $\text{RankMe} = \exp(-\sum_i p_i \log p_i) \in$

$[1, 32]$ and condition number $\kappa = \sigma_{\max}/\sigma_{\min} \geq 1$. Because $\mathbf{K}$ is initialized near the identity, the unregularized operator is already near full rank ($\mathrm{RankMe}(\mathbf{K}) = 31.53$, $\kappa = 1.437$); increasing λ_{ent} tightens conditioning monotonically ($\kappa = 1.255$ at the deployed $\lambda_{\text{ent}} = 0.50$, and $\kappa = 1.154$, $\mathrm{RankMe}(\mathbf{K}) = 31.95$ at $\lambda_{\text{ent}} = 2.00$) while lowering loss variation ($\mathrm{CV}_{\mathcal{L}} : 0.565 \rightarrow 0.545$). Because $\mathcal{L}_{\text{Lyap}}$ applies a stop-gradient to z_{ctx}, encoder effective ranks remain invariant across all strengths ($\mathrm{RankMe}(z_{\text{ctx}}) \in [5.16, 5.54]$, $\mathrm{RankMe}(z_{\text{tgt}}) \in [11.54, 11.58]$), as do false-alarm rates (0.016–0.017 causal, 0.070–0.074 buffered; buffered ER 0.250–0.266, F1 0.032–0.047; causal F1 0.004–0.011), while the deployed setting ($\lambda_{\text{ent}} = 0.50, \beta_{\text{norm}} = 1$) attains the highest causal event recall (0.444 vs. 0.369 unregularized). On an eight-stream decoupling panel, pairing detached covariance alignment ($\beta_{\text{Lyap}} = 0.02$) with the smooth Bregman barrier cuts causal detection delay from 127.0 steps (TS-JEPA) and 80.2 steps ($\beta_{\text{Lyap}} = 0$) to 58.2 steps while raising causal Point-F1 from 0.044 to 0.094.

C. Ablation 3: Fault-Class Breakdown, Quiet-Fault Mechanism, and Hybrid Monitor

Comparing the auxiliary context-variance input-channel gate against uniform channel weighting on four streams confirms that uniform weighting achieves higher macro Point-F1 (0.118 vs. 0.077), so the deployed pipeline retains uniform input encoding. Across six injected single-channel fault types on six multivariate streams (Fig. 3(b)), uniform latent encoding (`Unif.`) achieves 0.278 event recall on abrupt $+5\sigma$ spikes (0.306 with variance gating), versus 0.194 event recall and 0.020–0.030 point recall on persistent quiet faults (flatlines, stuck sensors, dropouts) and low-amplitude steps (0.043–0.068 under variance or median absolute deviation (MAD) gating). When a single channel k freezes or shifts by δ out of K channels, its contribution to the pre-LayerNorm pooled vector scales as $O(1/K)$ and is rescaled onto the $\sqrt{D}$ sphere by LayerNorm along with nominal cross-channel variance. Standalone sensor monitors, namely a rolling variance-collapse detector (`lowvar`) and maximum channel z-score (`dev_maxz`, 0.087–0.108 point and 0.250–0.306 event recall at 14.1% FPR), trigger on different streams than TS-JEPA. Consequently, a hybrid union detector (**Hybrid**, triggering when either TS-JEPA or `lowvar` alarms) raises point recall to 0.202–0.318 and event recall to 0.417–0.500 across all six fault classes at +15.1% additional nominal FPR (1.9% $\rightarrow$ 17.0%; or 0.333–0.472 event recall at 16.1% FPR with `dev_maxz`).

D. Ablation 4: Multiscale Dyadic-Shell Front-End

In Table II(b), replacing the standard convolutional stem with the 4-shell Fourier front-end increases macro PR-AUC from 0.213 to 0.263 (+0.050, led by SMD and GECCO) and lowers calibration residual kurtosis (1.409 $\rightarrow$ 1.069), while macro Point-F1 shifts from 0.118 to 0.078 due to conservative threshold tightening on SMAP `P-3`. The module thus acts as a ranking-conditioning stem rather than a uniform F1 gain.

E. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing the 14 factorial arms across all 19 non-GHL streams under identical splits, training budgets, and comparable parameter counts in Table II(c) isolates six architectural factors:

1) **Prediction vs. reconstruction in observation space (`TCN-ObsPred` vs. `TCN-ObsRecon`):** Using the exact same `HybridTCNEncoder` backbone, forecasting versus reconstructing raw observations leaves kurtosis ($\gamma_2 = 6.10$ vs. 6.24), calibration-to-holdout drift ($D_{\text{KS}} = 0.463$ vs. 0.445), and causal FPR (24.7% vs. 24.5%) virtually unchanged.
2) **LayerNorm bottleneck + saliency gate in observation space (`TCN-ObsPred-LN`):** Adding the $D = 32$ LayerNorm bottleneck and coordinate saliency gate to the observation forecaster (307k parameters) compresses tail kurtosis to $\gamma_2 = 0.91$ and halves causal FPR to 11.7%, but leaves calibration-to-holdout drift high ($D_{\text{KS}} = 0.376$) and lowers causal Point-F1 to 0.084.
3) **Learned latent target projection (`TS-JEPA`, `OpEntropy`, `Reynolds`):** Moving the prediction target into the learned $D = 32$ latent space (269k–282k online parameters) halves calibration-to-holdout drift relative to `TCN-ObsPred-LN` ($D_{\text{KS}} : 0.376 \rightarrow 0.193$–$0.274$, paired Wilcoxon $p \leq 0.001$), cuts causal FPR to 1.9%–2.8% for base predictors, and raises causal Point-F1 from 0.084 to 0.153–0.161.
4) **Learned TCN encoder vs. linear subspace (`PCA-LinPred-LN`) and isolation forest:** `PCA-LinPred-LN` shares $D = 32$, LayerNorm, and the saliency gate, achieving 2.1% causal FPR via tail bounding, but its causal Point-F1 collapses to 0.017 (0.104 buffered; `IsoForest` event recall 0.149), proving that the learned nonlinear encoder is required for anomaly discriminability.
5) **Temporal foresight (`JEPA-CtxRecon`), EMA target (`JEPA-NoEMA`), and saliency gate (`TS-JEPA-NoGate`):** Autoencoding context embeddings (`JEPA-CtxRecon`) reduces drift ($D_{\text{KS}} = 0.203$, causal FPR 1.7%) at lower causal F1 (0.112 vs. 0.161); hard-copy targets (`JEPA-NoEMA`) reach 0.9% causal FPR and 0.146 causal F1; and replacing the saliency gate with plain ℓ_2 norm (`TS-JEPA-NoGate`) yields 1.8% causal FPR and 0.156 causal F1, confirming that false-alarm reduction does not depend on the saliency gate.

VII. Results and Discussion

A. Benchmark Performance on Continuous Telemetry

Table III reports all 33 continuous telemetry streams under the chronological held-out SPOT protocol ($u = p_{98}, q = 10^{-3}$): Panel (a) uses retrospective buffered attribution, Panel (b) strictly causal streaming, and Panel (c) per-dataset breakdowns including anomaly event counts (E) and both causal and buffered held-out nominal FPRs. Evaluating realized FPR and out-of-sample calibration error $|\text{FPR}_{\text{hold}} - q|$ as primary outcomes shows a large relative reduction in false

TABLE III

HELD-OUT CALIBRATION BENCHMARK (SPOT, $u{=}p_{98}, q{=}10^{-3}$): (A) RETROSPECTIVE BUFFERED ($S{=}64$) AND (B) STRICTLY CAUSAL TRAILING ATTRIBUTION ACROSS 19 NON-GHL STREAMS AND 14 GHL STREAMS, (C) PER-DATASET BREAKDOWN (INCLUDING ANOMALY EVENT COUNTS E AND CAUSAL/BUFFERED HELD-OUT FPR) ACROSS $N{=}7$ TELEMETRY FAMILIES, AND (D) 130-STREAM SINGLE-SEED CROSS-CHECK. BEST IN **BOLD**, SECOND UNDERLINED.

	(a) Retrospective Buffered, Non-GHL (19 Streams)					(b) Strictly Causal Trailing, Non-GHL (19 Streams)				GHL Buffered (14)		Macro
Architecture	**FPR ↓**	**Point-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Point-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Point-F1 ↑**	**DS-F1 ↑**
Latent-space predictive architectures (JEPAs)												
TS-JEPA (unconstrained)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Operator-Entropy JEPA	0.101	**0.233**	**0.351**	0.550	79.5	0.028	0.153	0.517	112.5	0.012	0.001	**0.125**
Reynolds-Stress JEPA	0.190	0.148	0.285	0.421	116.5	0.086	0.060	0.474	133.9	0.009	0.002	0.073
Reynolds ($\alpha_{\text{stress}}{=}0$)	0.109	0.167	0.281	0.419	120.7	**0.019**	0.051	0.424	228.4	–	–	0.121
Observation-space reconstructive baselines												
TimesNet	0.300	0.197	0.329	**0.583**	**48.5**	0.243	**0.209**	**0.588**	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Non-neural linear baseline												
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Telemetry Family Descriptors					Buffered Point-F1 (Stream Mean) ↑					Causal / Buffered Held-Out Nominal FPR ↓				
Dataset	n	K	E	π	$\text{F1}_{\text{all-pos}}$	**TS-JEPA**	**OpEnt**	**Reynolds**	**TimesNet**	**TranAD**	**TS-JEPA**	**OpEnt**	**Reynolds**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.145	0.122	0.156	**0.229**	0.144	.030/.091	.018/.059	**.010/.039**	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.001	0.002	**0.022**	0.020	**.004/.004**	.009/.012	.006/.009	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	0.000	**0.008**	0.000	0.000	0.000	.004/.017	.024/.082	.012/.016	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	0.161	**0.323**	0.114	0.158	0.151	.054/**.095**	.069/.177	**.024**/.106	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.379	**0.396**	0.237	0.289	0.167	.014/.135	**.013/.125**	.226/.486	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	0.000	**0.001**	0.000	.008/.014	.008/.026	.008/.017	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	**0.119**	0.024	0.000	0.000	0.000	**.000/.000**	**.000/.000**	**.000/.000**	**.000/.000**	**.000/.000**

Note: Out-of-sample calibration error $|\text{FPR}_{\text{hold}} - q|$ at $q{=}0.001$ equals FPR $- 0.001$. 95% stream-bootstrap CIs on 19 non-GHL streams for Panel (a) Buffered FPR / Point-F1: TS-JEPA [.048, .145] / [.113, .319], OpEntropy [.043, .180] / [.131, .355], Reynolds [.092, .304] / [.069, .239], Reynolds ($\alpha_{\text{stress}}{=}0$) [.058, .166] / [.077, .276], TimesNet [.162, .446] / [.098, .315], TranAD [.177, .445] / [.053, .232], Linear AR [.001, .011] / [.012, .104]; for Panel (b) Causal FPR / Point-F1: TS-JEPA [.007, .046] / [.063, .287], OpEntropy [.007, .063] / [.070, .247], Reynolds [.035, .141] / [.028, .100], Reynolds ($\alpha_{\text{stress}}{=}0$) [.006, .038] / [.020, .094], TimesNet [.123, .375] / [.109, .326], TranAD [.119, .337] / [.058, .241]. In Panel (c), E is total test anomaly events ($34, 3, 2, 5, 7, 13, 3{,}137$ in the held-out half); per-dataset Causal / Buffered FPR stream SDs (and seed SDs) are: Daphnet .017–.047 / .061–.100 (JEPAs) vs. .337–.402 / .381–.430 (reconstructors; seed SD .005–.041); GHL .005–.014 / .006–.019 vs. .041–.065; MSL .047–.115 / .147–.224 vs. .228–.281; SMAP .007–.019 / .125–.139 (TS-JEPA/OpEnt) vs. .080–.150 (reconstructors). **(d) 130-Stream Single-Seed Cross-Check (Point-F1, dataset-macro / stream mean across 11 datasets):** NCAD $0.058/0.088$; DCdetector $0.070/0.097$; Anomaly Transformer $0.091/0.082$; TimesNet $0.129/0.151$; TranAD $0.156/0.152$; TS-JEPA **0.160 / 0.186**.

alarms and calibration error, whereas aggregate Point-F1 and PR-AUC remain comparable to reconstructive baselines.

Lower Realized FPR and Calibration Error on Held-Out Telemetry: On the 19 non-GHL streams (Table III(a),(b)), latent scoring lowers held-out nominal FPR and calibration error $|\text{FPR}_{\text{hold}} - q|$ three- to four-fold under buffered attribution (9.3% and 10.1% for TS-JEPA and OpEntropy JEPA versus 30.0% and 30.7% for TimesNet and TranAD) and eight- to ten-fold under causal streaming (2.4% and 2.8%, calibration error 0.023 and 0.027, versus 24.3% and 22.2%, calibration error 0.242 and 0.221). The reduction holds across all 33 streams in paired tests (buffered FPR -14.4% to -15.3% versus TimesNet, $p \leq 0.006$; 95% CI for OpEntropy $[-24.1\%, -6.5\%]$) and across the $N = 7$ dataset families ($p = 0.016$–0.031). Under zero-tuning LODO evaluation (where neither hyperparameter selection nor threshold calibration accesses the held-out dataset family), leaving one dataset out at a time keeps causal FPR at 1.0%–2.3% and buffered FPR at 3.8%–7.0% for JEPAs versus 12.1%–18.4% causal and 8.4%–22.1% buffered for reconstructors. In Panel (c), under strictly causal attribution, latent predictors achieve lower held-out FPR than TimesNet and TranAD on every multi-stream family including MSL (2.4%–6.9% vs. 13.2%–16.1%), whereas under buffered attribution TimesNet has a lower FPR than OpEntropy on MSL (13.3% vs. 17.7%), and on single-stream Genesis, Metro, and SWAN reconstructors emit zero alarms because extreme calibration tails set overly conservative thresholds.

Detection Quality Is Comparable, Not Superior: Lower false-alarm rates do not make latent scoring a universally stronger detector. Buffered Point-F1 and event recall for TS-JEPA and OpEntropy JEPA fall in the same range as TimesNet and TranAD (Table III(a)), and paired differences across all 33 streams are indistinguishable from zero (Point-F1 $+0.012$ and $+0.051$, $p = 0.54$ and 0.88; PR-AUC -0.000 and $+0.023$, $p = 0.39$ and 0.16 versus TimesNet and TranAD). Per-dataset standard deviations are as large as the means (Table III(c)). Only SMAP beats the trivial always-alarm F1 of 0.236 (0.379 and 0.396), although PR-AUC exceeds base prevalence π on both SMAP (0.504–0.525 vs. $\pi = 0.150$) and MSL (0.422–0.434 vs. $\pi = 0.255$).

Domain and Operational Applicability Boundaries: Synthesizing performance across fault classes and sampling rates defines four operational regimes:

1) *Favorable operating regime:* Multi-channel spacecraft and wearable kinematic arrays ($K \geq 9$, SMAP, MSL, Daphnet) with persistent localized transients, where suppressing nuisance false alarms under automated calibration is the primary operational requirement.
2) *Latency-constrained telemetry:* Systems where a 63-step emission lag is prohibitive for immediate tripwire response (such as hourly Metro where 63 steps $= 63$ h), requiring short horizons $S \in \{1, 4\}$ (Section VII-D).
3) *Quiet single-channel faults:* Inactive sensor modes (flatlines, dropouts) that LayerNorm embeddings absorb as nominal background, requiring the hybrid union detector (`TS-JEPA ∨ lowvar`) when single-channel coverage is mandatory (Section VI-C).

4) *Continuous non-stationary process ramps:* Industrial heating loops and traffic drifts (GHL, Metro; 0.000–0.022 F1), which defeat both static and rolling thresholds.

B. Evaluating Physics-Inspired Structures: Component-Level Justification

Rather than treating operator rank or contractivity as proxies for Point-F1, Tables II(a),(c) and III(a),(b) and Fig. 3(a) isolate three component-level effects beyond unconstrained `TS-JEPA`: (i) *Koopman spectral entropy and Bregman barrier (`OpEntropy JEPA`)* tighten operator conditioning ($\kappa(\mathbf{K})$: $1.437 \rightarrow 1.255$, $\mathrm{RankMe}(\mathbf{K})$: $31.53 \rightarrow 31.85$) and shorten mean buffered detection delay from 110.3 to 79.5 steps ($124.0 \rightarrow 112.5$ causally across 19 streams, and $127.0 \rightarrow 58.2$ on the decoupling panel), while leaving Point-F1 equivalent down to $\Delta = \pm 0.015$ ($p_{\mathrm{TOST}} = 0.0024$, 90% CI $[-0.0112, +0.0029]$); (ii) *Cayley-accretive Crouzeix transport (`Reynolds` with $\alpha_{stress} = 0$)* guarantees unit-disk stability ($W(\mathbf{W}_{\mathrm{Crouzeix}}) \subset \mathbb{D}$), achieves the lowest drift ($D_{\mathrm{KS}} = 0.193$) and causal held-out FPR (1.9% across 19 streams), and attains the highest matched-FPR causal event recall (0.457 at f^*=0.01 and 0.633 at f^*=0.05, Table II(c)); and (iii) *second-moment Stiefel stress closure ($\alpha_{stress} = 0.1$)* tracks subgrid covariance shifts on $\mathcal{S}_+^{16}$, but adding Frobenius covariance discrepancy to the score inflates causal FPR from 1.9% to 8.6% and lowers Point-F1 ($0.167 \rightarrow 0.148$ across all seeds; $0.167 \rightarrow 0.158$ on matched seeds, 90% CI $[-0.052, +0.032]$), which we report as a negative aggregate detection result.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Comparing Panels (a) and (b) of Table III isolates the cost of zero-lookahead streaming. Without retrospective smoothing, lower false-alarm rates (2.4%–2.8%) trade onset speed and sensitivity: causal Point-F1 and event recall are more conservative (0.153–0.161 and 0.487–0.517, vs. 0.209 and 0.588 for TimesNet), and mean delay increases to 112.5–124.0 steps (62.2 for TimesNet). On five non-stationary incident streams with discrete regime shifts or localized variance steps (including Room-Occupancy, CalIt2, and SMAP `P-3`), pairing latent scores with dynamic rolling SPOT suppresses false alarms by up to 21×. However, as shown in Table IV(b), this suppression applies specifically to discrete shifts; under continuous monotonic mean ramps, trailing calibration windows continually lag the drifting mean, producing high false-alarm rates (32.6%–38.5%) across all models.

D. Horizon Length, Score-Emission Latency, and False-Alarm Control

Target horizon $S = 64$ imposes a 63-step emission lag before an alarm can trigger (Section IV-C). To evaluate detection quality versus latency against baselines at matched emission lag $S - 1 \in \{0, 3, 15, 31, 63\}$, Fig. 4(a) retrains models across $S \in \{1, 4, 16, 32, 64\}$ on five streams (three seeds). At $S = 1$ (zero lag) and $S = 4$ (3-step lag), OpEntropy JEPA holds causal FPR to 0.5% and 0.7% (0.4% and 0.6% for TS-JEPA, vs. 2.5% and 3.1% for TimesNet) while achieving higher causal PR-AUC (0.182 and 0.189 vs. 0.150 and 0.162 for TimesNet), comparable Point-F1 at $S = 1$ (0.028 vs. 0.025), and event recall of 0.729 and 0.624 ($\Delta t = 76.3$ and 179.3 steps; TS-JEPA $\Delta t = 57.8$ and 85.2 steps). As S increases across $\{16, 32, 64\}$, TimesNet causal FPR escalates nearly nine-fold ($6.1\% \rightarrow 9.8\% \rightarrow 21.3\%$), whereas OpEntropy ($0.3\% \rightarrow 1.1\% \rightarrow 2.0\%$) and TS-JEPA ($0.3\% \rightarrow 0.3\% \rightarrow 0.4\%$) remain bounded. Thus $S = 64$ suits high-rate sensors (≥ 1 Hz, e.g., 64 Hz Daphnet where 63 steps = 0.98 s) or slow subsystem trend monitoring (0.1 Hz SMAP/MSL/GHL, 10.5 min), whereas $S = 64$ is unacceptable for low-rate infrastructure streams ($\leq 1\,\mathrm{min}^{-1}$, e.g., hourly Metro where 63 h lag is prohibitive) or rapid tripwires, which require $S \in \{1, 4\}$.

E. Deployable Calibration Quantiles vs. Matched Realized Test-FPR Operating Points

Because reconstruction baselines calibrate to 22%–31% realized FPR versus 1%–10% for latent predictors, and long segments (402 steps on Daphnet, 1,102 on SMAP) allow a 30% random-alarm detector to touch > 95% of segments via $1 - (1 - \mathrm{FPR})^L$, we compare models under two operating regimes:

1) **Deployable nominal calibration quantiles (q^*, Fig. 4(b)):** At causal calibration quantile $q^* = 0.050$ (set without test labels), TS-JEPA and OpEntropy JEPA achieve higher event recall (0.730 and 0.752) and precision (0.251 and 0.259) at fewer false alarms (177–203 per 1,000 nominal steps) than TimesNet (0.618 recall, 262 alarms) and TranAD (0.602 recall, 270 alarms).
2) **Post-hoc matched realized test-FPR budgets (f^*, Table II(c)):** Equalizing realized test nominal FPR across models, at $f^* = 0.01$ latent models nearly double causal event recall (0.371–0.457) compared to TimesNet (0.217) and TranAD (0.149), whereas at relaxed budgets ($f^* \geq 0.10$) recall curves converge ($p = 0.625$). Latent scoring thus specifically improves sensitivity in the low-FPR regime ($f^* \leq 0.01$).

F. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Static extreme-value thresholding does not achieve its nominal design risk $q = 10^{-3}$ (0.1%) out of sample. Across all 33 streams (Table IV(a)), standard SPOT achieves stream-macro causal FPR of 1.51% for TS-JEPA and 1.95% for OpEntropy ($|\mathrm{FPR}_{\mathrm{hold}} - q| = 0.014$ and 0.019, corresponding to 15× and 20× the $q = 10^{-3}$ design risk), and buffered FPR of 5.54% and 6.38% (55–64×). These rates are an order of magnitude below TimesNet (16.11% causal, $|\mathrm{FPR}_{\mathrm{hold}} - q| = 0.160$, 161×) and TranAD (14.23% causal, $|\mathrm{FPR}_{\mathrm{hold}} - q| = 0.141$, 142×), showing that latent prediction reduces false alarms and drift without solving static calibration at $q = 10^{-3}$. Comparing static SPOT against runs-declustered GPD fitting (`Decl. SPOT`, run-length $r = 5$ via the Ferro–Segers extremal index [30]) and adaptive rolling calibrators (`Roll. emp.`, `Roll. SPOT`) across 16 streams and nine

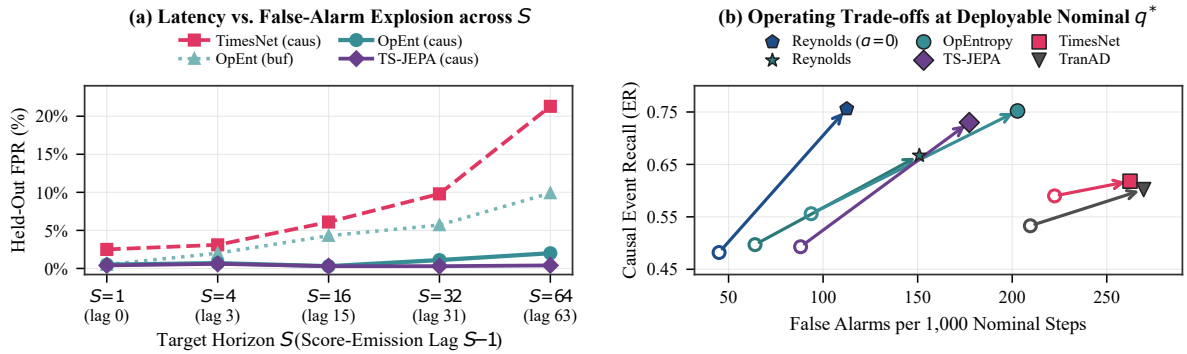


Fig. 4. Horizon length and deployable calibration trade-offs. **(a)** Held-out FPR and event recall across target horizons $S \in \{1, 4, 16, 32, 64\}$ at matched score-emission latency $S-1 \in \{0, 3, 15, 31, 63\}$ steps (five streams, three seeds). **(b)** Causal event recall versus false alarms across 19 non-GHL streams as nominal calibration quantiles shift from $q^*=10^{-3}$ (open circles) to $q^*=0.05$ (filled markers).

TABLE IV
THRESHOLD CALIBRATOR COMPARISON AT DESIGN RISK $q=10^{-3}$: (A) STATIC RULES ACROSS 33 STREAMS (CELL-WEIGHTED MEANS, STREAM-MACRO IN PARENTHESES, AND CALIBRATION ERROR $|\text{FPR}_{\text{HOLD}}-q|$), (B) CONTROLLED-DRIFT STRESS TEST INCLUDING RUNS-DECLUSTERED SPOT ($n=16$), AND (C) THREE-STAGE REFITTED PERMUTATION DECOMPOSITION OF EXCESS CAUSAL FPR ($n=18$).

(a) Model	Calibrator Rule	Causal FPR	$\lvert\text{FPR}-q\rvert$ (Over.)	Buffered FPR	F1 (buf)
TS-JEPA	Standard SPOT	0.017 (0.015)	.014 (15×)	0.064 (0.055)	0.1386
	SPOT + Kurtosis	0.019 (0.016)	.015 (16×)	0.163 (0.140)	0.1498
	Empirical p99.5	0.025 (0.023)	.022 (23×)	0.076 (0.066)	0.1399
	Validation Max	0.016 (0.014)	.013 (14×)	0.066 (0.057)	0.1385
OpEntropy	Standard SPOT	0.021 (0.020)	.019 (20×)	0.073 (0.064)	0.1573
	SPOT + Kurtosis	0.021 (0.019)	.018 (19×)	0.135 (0.117)	0.1610
	Empirical p99.5	0.030 (0.028)	.027 (28×)	0.088 (0.078)	0.1633
	Validation Max	0.019 (0.017)	.016 (17×)	0.073 (0.064)	0.1566
Reynolds	Standard SPOT	0.060 (0.052)	.051 (52×)	0.131 (0.113)	0.1003
	SPOT + Kurtosis	0.056 (0.048)	.047 (48×)	0.179 (0.153)	0.0987
	Empirical p99.5	0.069 (0.061)	.060 (61×)	0.143 (0.125)	0.1027
	Validation Max	0.060 (0.052)	.051 (52×)	0.132 (0.114)	0.0968
TimesNet	Standard SPOT	0.180 (0.161)	.160 (161×)	0.222 (0.199)	0.1395
	SPOT + Kurtosis	0.184 (0.162)	.161 (162×)	0.228 (0.201)	0.1352
	Empirical p99.5	0.191 (0.175)	.174 (175×)	0.234 (0.213)	0.1413
	Validation Max	0.179 (0.160)	.159 (160×)	0.226 (0.203)	0.1410
TranAD	Standard SPOT	0.160 (0.142)	.141 (142×)	0.222 (0.196)	0.0936
	SPOT + Kurtosis	0.172 (0.151)	.150 (151×)	0.245 (0.215)	0.0940
	Empirical p99.5	0.171 (0.154)	.153 (154×)	0.234 (0.209)	0.0962
	Validation Max	0.158 (0.139)	.138 (139×)	0.227 (0.201)	0.0952

(b) Drift condition	Stat. SPOT	Decl. SPOT	Stat. emp.	Roll. emp.	Roll. SPOT
Stationary (no drift)	0.061	0.159	0.162	0.016	0.047
Mean shift $+1.0\sigma$	0.087	0.209	0.216	0.011	0.043
Regime change $+1\sigma$	0.073	0.182	0.187	0.015	0.070
Variance scale 2.0	0.082	0.186	0.195	0.016	0.047
Autocorrelation 0.9	0.113	0.218	0.232	0.024	0.055
Continuous mean ramp	0.075	0.182	0.186	0.326	0.385

	Refitted Stage Causal FPR			Share of Excess (%)		
(c) Model	FPR_{chr}	FPR_{blk}	FPR_{pt}	Inter-Blk	Serial	i.i.d.
TS-JEPA	0.0121	0.0058	0.0019	56.6	35.3	8.1
OpEntropy	0.0185	0.0073	0.0022	64.1	29.1	6.9
Reynolds	0.0022	0.0037	0.0030	0.0	25.9	74.1
TimesNet	0.0872	0.0204	0.0037	77.5	19.4	3.1
TranAD	0.0741	0.0067	0.0032	92.3	4.7	3.0

In (a), Over. is the overshoot ratio $\text{FPR}_{\text{macro}}/q$ at $q=10^{-3}$. In (b), causal TS-JEPA / OpEntropy achieve .0044/.0046 stationary FPR under Decl. SPOT ($r=5$, vs. .0072/.0043 Stat. SPOT) and .0006–.0040 under Roll. emp. across step/variance/regime shifts (vs. .008–.030 for TimesNet). In (c), 95% CIs for $\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$: TS-JEPA $+.0063$ $[-.005, +.019]$, OpEnt $+.0112$ $[-.002, +.034]$, Reynolds $-.0015$ $[-.005, +.002]$, TimesNet $+.0668$ $[+.031, +.108]$, TranAD $+.0674$ $[+.035, +.106]$.

drift regimes in Table IV(b) shows that: (i) on stationary nominal streams under causal scoring, `Decl. SPOT` lowers TS-JEPA FPR from 0.72% to 0.44% (1.36% → 1.07% on the 4-stream tail panel, and 24.4% → 18.9% for TimesNet), whereas under buffered smoothing declustering reduces N_u exceedances to $N_c \approx$ 4–6 clusters, increasing GPD variance (0.159 pooled); and (ii) adaptive `Roll. emp.` reduces pooled FPR from 6.1%–11.3% to 1.1%–2.4% under discrete step, variance, and autocorrelation shifts, holding causal OpEntropy (0.06%–0.10%) and TS-JEPA (0.40%) near the 0.1% design risk, whereas continuous monotonic ramps defeat trailing rolling windows (32.6%–38.5%). Three diagnostics explain the static overshoot:

1) **GPD tail stability and quantile/length sensitivity:** On the 19 non-GHL streams at $u = p_{98}$, median exceedance count is $N_u = 28$ causally (21 buffered) with 0% solver fallback and low threshold uncertainty ($\text{SE}(\tau)/\tau = 2.6\%$–$3.5\%$ causal). Causally, Grimshaw GPD fits hit the lower shape boundary ($\xi = -0.90$) in 4.4% of JEPA cells (median $\xi = -0.27$) versus 34.4% for TimesNet and 54.4% for TranAD. Varying $u \in \{p_{95}, p_{98}, p_{99}\}$ leaves causal FPR unchanged (2.41%–2.42% for TS-JEPA), whereas halving calibration length (50% N_{cal}) raises causal FPR to 3.72% (TS-JEPA) and 4.13% (OpEntropy).
2) **Platykurtic collapse under total-threshold kurtosis scaling:** Total-threshold scaling $\tau_{\text{eff}} = \tau[1 + \frac{1}{6}\tanh(\gamma_2)]$ reduces buffered FPR on leptokurtic cells ($\gamma_2 \geq 0$, 4.49% → 1.11% for TS-JEPA), but 64-step buffered smoothing renders 31.9% of TS-JEPA cells platykurtic ($\gamma_2 < 0$). Scaling total τ pulls τ_{eff} below $u = p_{98}$ in 28.4% of cells, inflating buffered FPR from 6.40% to 16.30% (Table IV(a)). Excess-only scaling $\tau_{\text{exc}} = u + (\tau - u)[1 + \frac{1}{6}\tanh(\gamma_2)] \geq u$ prevents collapse below u (9.19% buffered, 2.40% causal).
3) **Three-stage refitted permutation decomposition (Table IV(c)):** Pooling nominal calibration and evaluation scores $\mathbf{v}_{\text{pool}} = [v_{\text{cal}}, v_{\text{hold}}]$ and *refitting* SPOT ($u = p_{98}, q = 10^{-3}$) on each permuted split separates chronological (FPR_{chr}), $L = 64$ block-permuted (FPR_{blk}, equalizing macro-block distributions while preserving intra-block autocorrelation), and pointwise i.i.d. permuted (FPR_{pt}) rates. Across the 18 streams with $N_{\text{cal}}, N_{\text{hold}} \geq 100$ (Table IV(c)), inter-block drift ($\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$) accounts for 56.6%–64.1% of excess causal overshoot for JEPAs (77.5%–92.3% for reconstructors) as descriptive point-estimate shares under the permutation protocol, while serial dependence ($\text{FPR}_{\text{blk}} - \text{FPR}_{\text{pt}}$) accounts for 29.1%–35.3%. Whereas reconstructors exhibit statistically resolved positive inter-block excess ($+0.067$ $[+0.031, +0.108]$ for TimesNet and $+0.067$ $[+0.035, +0.106]$ for TranAD), JEPA bootstrap intervals span zero ($[-0.005, +0.019]$ and $[-0.002, +0.034]$) because 11 of 18 streams achieve $\text{FPR}_{\text{chr}} = 0$.

VIII. Conclusion and Future Work

We evaluated how Joint-Embedding Predictive Architectures condition anomaly residuals under automated extreme-value calibration across 33 continuous telemetry streams under a chronological held-out protocol:

1) **Factorial Mechanism Isolation and Regularizer Equivalence:** Matched-encoder factorial ablations (269k–307k online parameters) decouple score compression from representation learning: LayerNorm bottlenecks bound residual kurtosis (γ_2 : $6.10 \rightarrow 0.91$), while learned latent projections halve calibration-to-holdout drift (D_{KS} : $0.376 \rightarrow 0.193$–0.214) and cut causal false-positive rates to 1.9%–2.8%. On a fixed Koopman backbone, the spectral regularizer satisfies TOST equivalence down to $\Delta = \pm 0.015$ ($p_{\mathrm{TOST}} = 0.0024$), OpEntropy shortens detection delay ($110.3 \rightarrow 79.5$ buffered; $127.0 \rightarrow 58.2$ causally on the decoupling panel), and short horizons ($S \in \{1, 4\}$) remove the 63-step lag.
2) **Relative False-Alarm Control, Detection Parity, and Hybrid Quiet-Fault Recovery:** Latent scoring lowers held-out false alarms and calibration error three-fold under buffered attribution and eight- to ten-fold under causal streaming, doubling causal event recall at 1% false-alarm budgets ($f^* = 0.01$) while aggregate Point-F1 and PR-AUC match reconstructive baselines. Pairing TS-JEPA with a rolling variance monitor raises quiet single-channel event recall from 0.19–0.28 to 0.42–0.50 at a $+15.1\%$ nominal FPR cost.
3) **Calibration Drift and Adaptive Thresholding:** Static SPOT overshoots its 0.1% design risk by 15–$64\times$ out of sample ($|\mathrm{FPR}_{\mathrm{hold}} - q| = 0.014$–$0.063$), driven by inter-block drift (56.6%–64.1% descriptive share) and serial dependence (29.1%–35.3%). Runs-declustered SPOT (0.44% stationary causal FPR) and rolling quantiles (0.06%–0.40% across step/variance shifts) narrow this gap, motivating future regime-conditioned conformal calibration.

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